

G2-02 Literature Review

1. NeRV: Neural Reflectance and Visibility Fields for Relighting and View Synthesis

This paper talks about learning a 3D representation of a scene that can still work when the lighting changes. One of the main things it learns is visibility in different directions, as well as information about the first surface a ray would hit. This is useful because the model is not only learning the final appearance of the scene, but also some of the physical information that causes that appearance.

I think this connects closely to G2-02 because we could also treat visibility as something the model directly predicts. For each local region and light source, the model could predict whether the light is visible and then possibly extend that to predict which object is actually blocking it. The main idea I would take away from this paper is that visibility should probably be an explicit part of the model instead of something hidden inside a larger network

2. Learning Visibility Field for Detailed 3D Human Reconstruction and Relighting

This paper talks about learning a visibility field while also making sure that the visibility predictions stay consistent with the geometry of the object. Even though the paper focuses on human reconstruction, the idea is still relevant because visibility is being learned as its own representation instead of only being used indirectly to produce the final image.

The part I found most useful for G2-02 is the idea of adding a physical or geometric constraint to the model. For example, if our model predicts that a certain building is responsible for blocking a light, that building should actually be in a position where it could block the ray. This could help make sure that the model is learning the correct reason for a shadow instead of just finding patterns that happen to give the right answer.

3. Neural Visibility Cache for Real-Time Light Sampling

This paper talks about using a neural network to predict visibility between points in a scene and different light sources. The main goal of the paper is to make rendering faster when there are many lights, since checking visibility for every point and every light can be expensive.

I think this is useful because our problem involves multiple light sources. One simple way to approach this could be to handle each light separately. For a local region and one light, the model could predict whether that light reaches the region and which object is most responsible for blocking it. Then the predictions from all of the lights could be combined to estimate the final illumination. This way seems simpler than giving every light to one large model at the same time

Thoughts on Possible Method

Based on these papers, I think a good starting point would be to handle each light separately and use the scene geometry to narrow down which objects could realistically block it. The model could then predict whether the light reaches the local region and which object is the main occluder. I would also add a simple physical consistency check so that if the model says an object is blocking the light, the geometry should support that. Then, later on, we can extend this to predict how close the current setup is to a visibility change, which I think would connect more to the event-boundary part of G2-02.

References

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